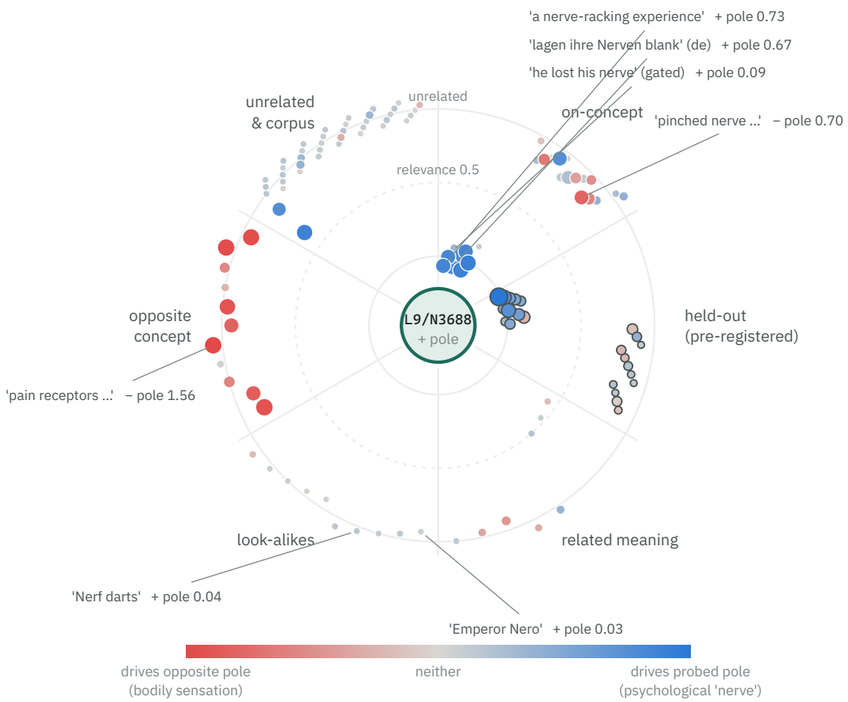
# Reading one neuron three ways

Layer 9, neuron 3688 of Prism ML's 1-bit Bonsai-8B is a context-gated axis for the word 'nerve': psychological strain and composure at one pole, bodily sensation at the other.

Forward: what text fires it. Inverse: solvers compose natural passages that hit requested activations. Induce: push it inside the model. Every number below is measured; predictions were committed to git before each test.

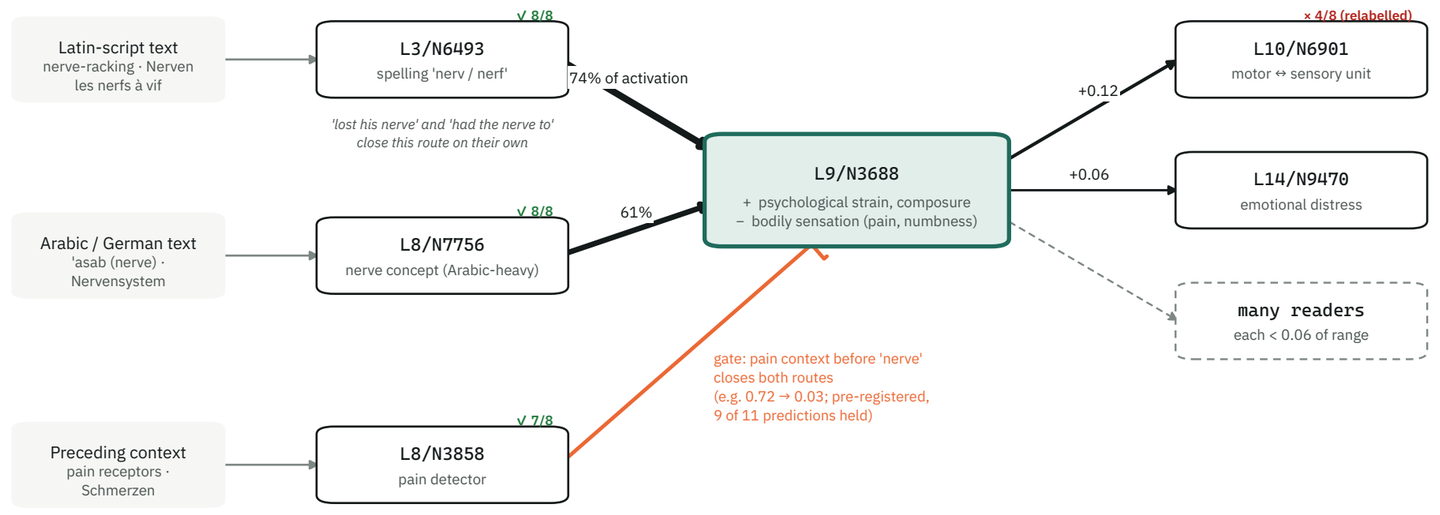
## A Every probe sentence around the neuron

angle: sentence type · radius: relevance to the label · colour: which pole it drives · size: strength




## B Its circuit: two input routes, a pain gate, a thin fan-out

causal effects from zeroing single neurons on 12 branch examples; labels checked by pre-registered tests

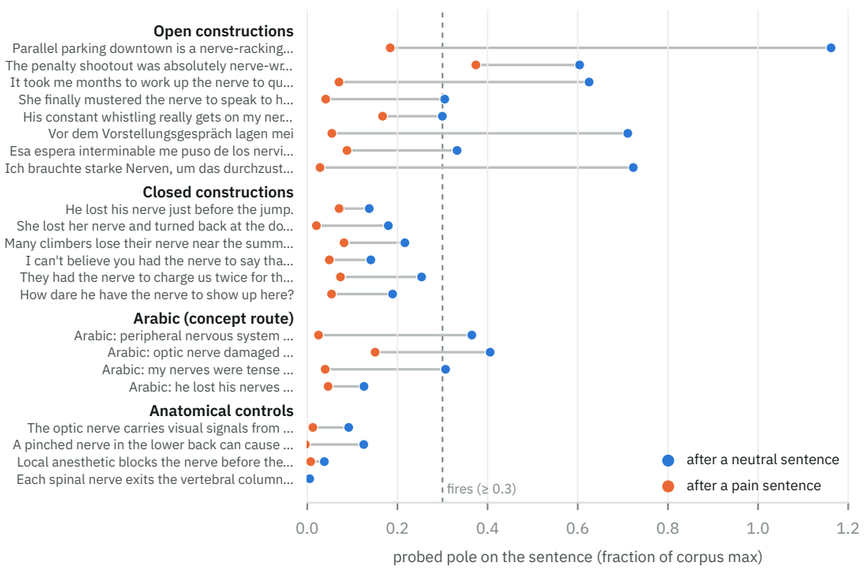



Input arrows: share of the probed pole's activation lost when that one neuron is zeroed. Output arrows: change in the reader (fraction of its range) when the probed neuron is zeroed.

Badges: pre-registered 8-sentence quick checks of each neighbour's label (fired / 8). A third, minor route ('strong') uses none of these inputs.

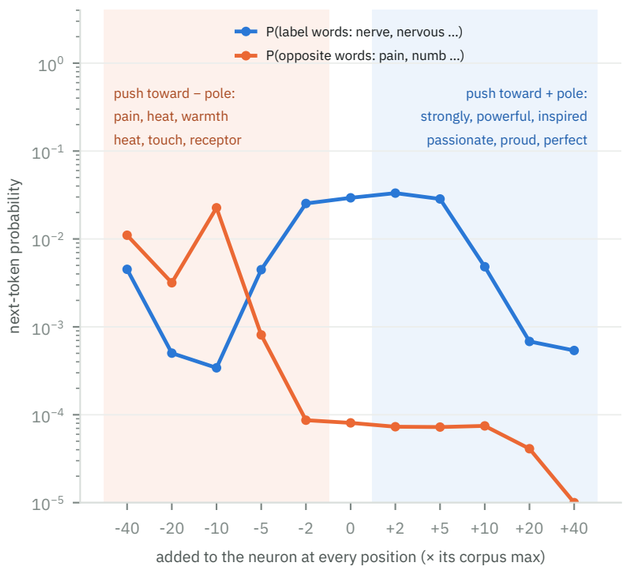
## C Pre-registered gate test

same sentence after a neutral vs a pain sentence



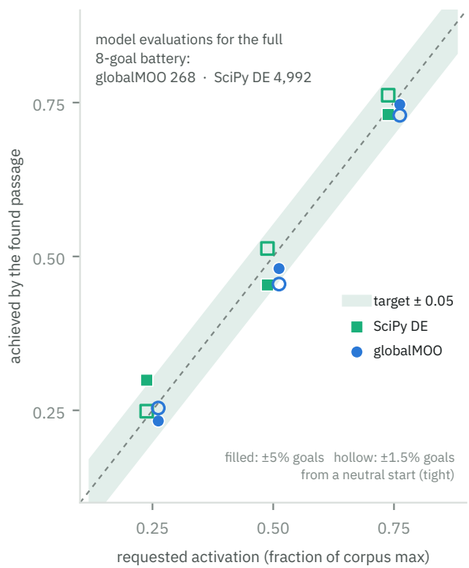

## D Induce: push the neuron, read the model

"Before the big presentation, I felt ..." 0: "nervous" · −5×: "the pain"



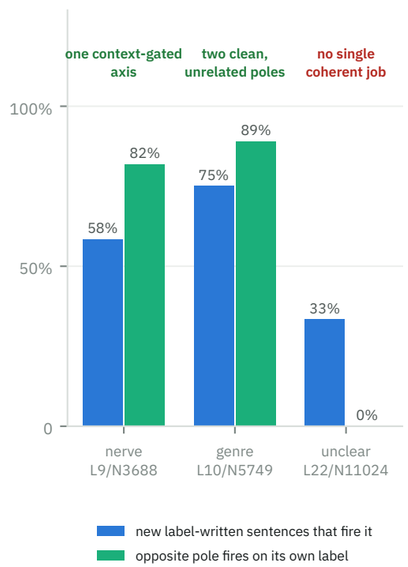

## E Inverse round trip

ask for a level, get a passage




## F Calibration

same probe, three neurons




Bonsai-8B (1-bit weights, Qwen3-8B architecture), run exactly from unpacked weights; neuron = one channel of silu(gate)·up; activations are fractions of each pole's maximum over an 8M-token corpus, read after neutral preambles. Inverse solvers: globalMOO (one model per neuron) and SciPy differential evolution over a bank of natural sentences.